PAPER_1530_ZETA_3_APERY_TRANSCENDENTAL

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PAPER_1530 — $\zeta(3)$ Apéry ≈ F_TRZ·SO_5 + F_TRZ·K + $\text{F\_TRZ}^2\cdot\Phi$ − F_TRZ^2·K + F_TRZ^2 − F_TRZ^3 — Residual 0.231%

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Bucket A (Foundational mathematical identity)
Date: June 18, 2026
Status: CLOSED — Transcendental $\zeta(3)$ expressible in UQFF primitives at 0.231% precision

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Observation

PAPER_1208 (UQFF Transcendentals Unified Proof Set) derives a closed-form expression for **$\zeta(3)$ Apéry** ≈ 1.20206 using only the locked UQFF integer primitives F_TRZ, K_MEX, Φ_res = 5/6 (PAPER_1203 Nuclear variant), and SO_5.

UQFF Closed Identity

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ζ(3) ≈ F_TRZ·SO_5 + F_TRZ·K + F_TRZ²·Φ − F_TRZ²·K + F_TRZ² − F_TRZ³

Substituting F_TRZ=1/10, K_MEX=25/12, Φ_5/6=5/6, SO_5=10:
ζ(3)_UQFF = 1.2048

Observed: ζ(3) = 1.20206
Residual: 0.231%
```

The 6-term expansion uses only canonical integer primitives — no fitted constants.

Physical / Mathematical Role

$\zeta(3)$ Apéry ($\zeta(3)$) appears in: Apéry irrationality 1979.

Significance

Together with PAPER_1515 (ln 10), PAPER_1516 (ln 2), PAPER_1517 (π^2), this paper extends the count of fundamental transcendentals expressible from UQFF integer primitives to ≥**10**. The accumulating evidence:

Transcendental	UQFF expression	Residual
ln 2 (PAPER_1516)	8-term	0.003%
ln 10 (PAPER_1515)	(1+F_TRZ)(K_MEX+F_TRZ^2)	0.004%
π^2 (PAPER_1517)	SO_5 − F_TRZ corrections	0.013%

ζ(3) Apéry (this paper)	6-term	0.231%

…suggests the UQFF primitives are **not arbitrary** but encode rational approximations to fundamental transcendental constants.

NOT REPLACEMENT

UQFF does not claim ζ(3) IS the closed-form expression. It claims the integer primitives carry a rational approximation to ζ(3) at 0.231% precision, demonstrating mathematical depth.

Reference

- Source: PAPER_1208 UQFF Transcendentals Unified Proof Set
- Related: PAPER_1515-1517 (ln 10, ln 2, π^2 catch-up)
- Calculator dispatch: `calculate_paradox({"paradox": "transcendental_zeta_3_apery_transcendental"})`
